# 1. Coordination and Response

## 1.1. Mammalian Nervous System

- The mammalian nervous system consists of two parts:
  - **Central Nervous System** (CNS) consists of the brain and spinal cord, which are the areas of coordination.
  - **The peripheral nervous system** (PNS) is comprised of nerves and neurones that coordinate and regulate body functions.
- Electrical impulses travel through the neurones.
- The nervous system helps with the coordination and regulation of body functions.

## 1.2. Types of Neurones

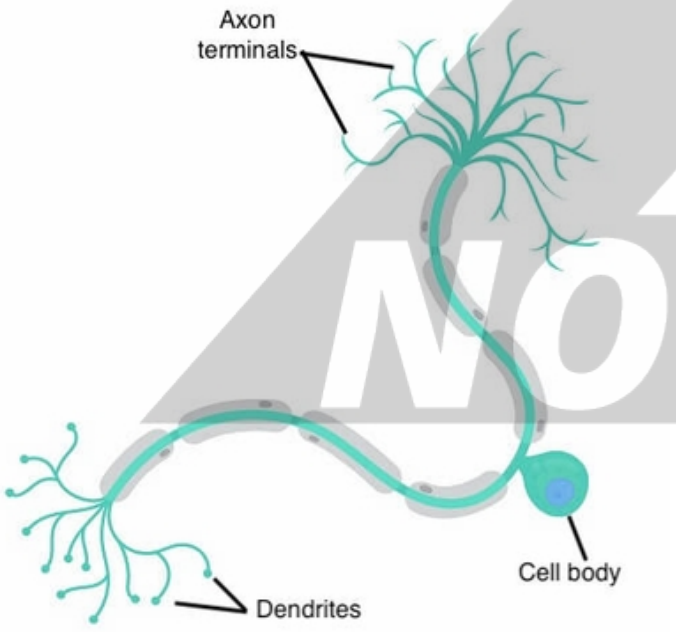
**Nerve Impulse:** an electrical signal that passes along the nerve cells called **neurones**

**Axons**: Ensure less time is wasted when the impulse is transferred from one neurone to another

**Dendrites**: Extensions which form a network for easy communication.

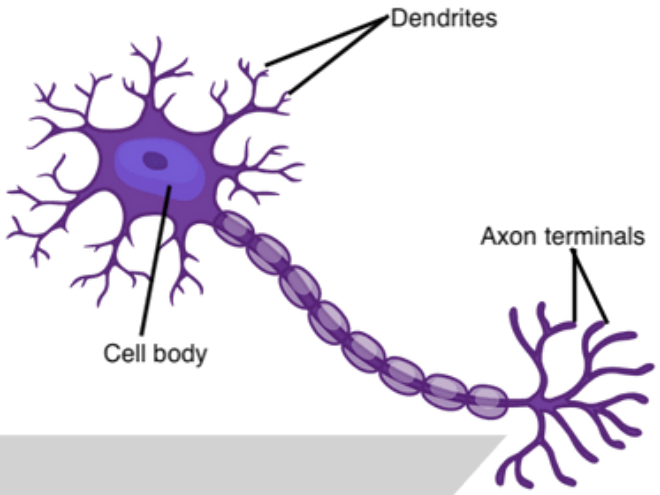
### Sensory Neurone

- Carry impulses from the **sense organs (sensory receptors) to the CNS**.



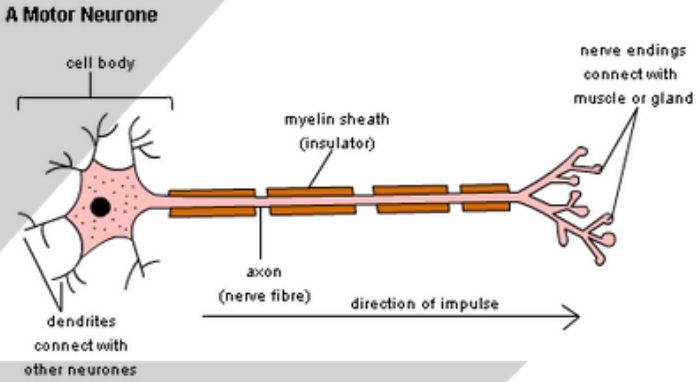

### Relay Neurone (Interneurone)

- The middleman inside the CNS connects sensory to motor neurones.




### Motor Neurone

- Carry impulses from the **CNS to effectors** (muscles or the glands)




## 1.3. Simple Reflex Arc

**Reflex Action:** automatically and rapidly integrates and coordinates the stimuli with the responses of effectors (muscles and glands).

- E.g. quickly removing your hand from the hot metal surface
- They involve three neurones: a sensory neurone, a relay neurone and a motor neurone.
- The gap between neurones is called a synapse.



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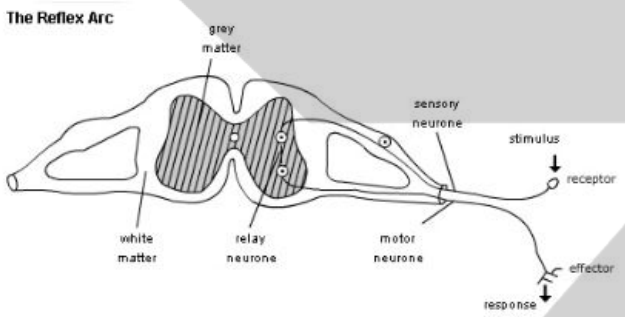
- How the simple reflex arc works:
  - A stimulus affects a receptor (cell or organ that converts a stimulus into an electrical impulse)
  - A sensory neurone carries impulses from the receptor to the CNS (spinal cord)
  - Connector/relay neurone carries impulse slowly (because it has no myelin sheath) across the spinal cord
  - The motor neurone carries impulses from the CNS to the effector
  - The effector (either a muscle or a gland) carries out the response

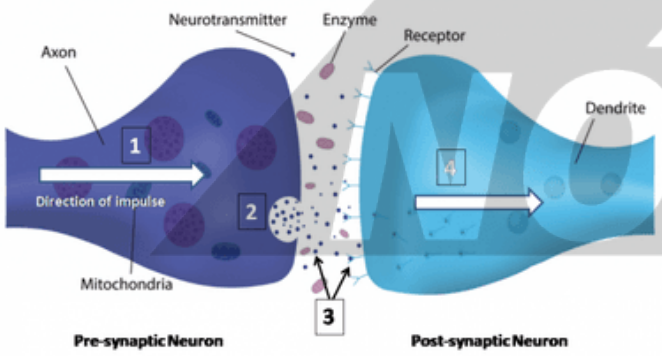
**Flowchart of the reflex arc:**

[Stimulus] → [Receptor (eg. in skin)] → [Sensory Neuron] → [Relay Neuron (in spinal cord)] → [Motor Neuron] → [Effector (e.g. Muscle)] → [Response (e.g. Pulling hand away)]




## 1.5. Synapse

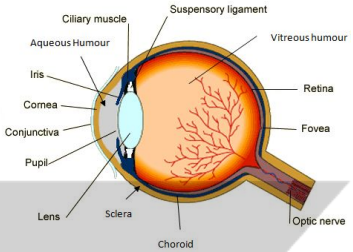
**Synapse:** a junction between two neurones, consisting of a gap across which impulses pass by diffusion of a neurotransmitter




- The synapses ensure that impulses travel in **one direction only**.
- **Synaptic cleft:** the small gap between each pair of neurones
- Inside the neurone's axon, there are 100s of tiny vacuoles (**vesicles**, each containing a chemical called neurotransmitter)
- When an impulse arrives, the vesicles move to the cell membrane and empty their content into the synaptic cleft.
- The neurotransmitter quickly diffuses across the tiny gap and attaches to receptor molecules in the cell membrane of the relay neurone.
- This can happen because the neurotransmitter molecules' shape complements the receptor shape.

## 1.7. Sense Organs

**Sense Organ:** groups of receptor cells responding to specific stimuli: light, sound, touch, temperature, and chemicals.




- **Cornea:** refracts light
- **Iris:** controls how much light enters the pupil
- **Lens:** focuses light onto the retina
- **Retina:** contains light receptors, some sensitive to light of different colours (Rods and cones)
- **Optic nerves:** carry impulses to the brain
- **Pupil**: A hole which allows light to enter

## 1.8. Pupil Reflex

- Adjusting for high and low light intensity
- An involuntary response

| | Low Light Intensity | High Light Intensity |
|---|---|---|
| Radial Muscles | Contract | Relax |
| Circular Muscles | Relax | Contract |
| Pupil Size | Wide | Narrow |
| Amount of Light Entering | More | Less |

## 1.9. Accommodation

**Accommodation**: Adjusting for near and distant objects.

- The way the **lens** brings about **fine focusing** is called **accommodation**
- The lens is **elastic,** and its shape can be changed when the **suspensory ligaments** attached to it become **tight** or **loose**
- The contraction or relaxation of the ciliary muscles brings about the changes

| | Near Object | Distant Object |
|---|---|---|
| Ciliary muscles | Contract | Relax |
| Suspensory Ligaments | Slack | Tighten |
| Lens | Short and fat | Long and thin |



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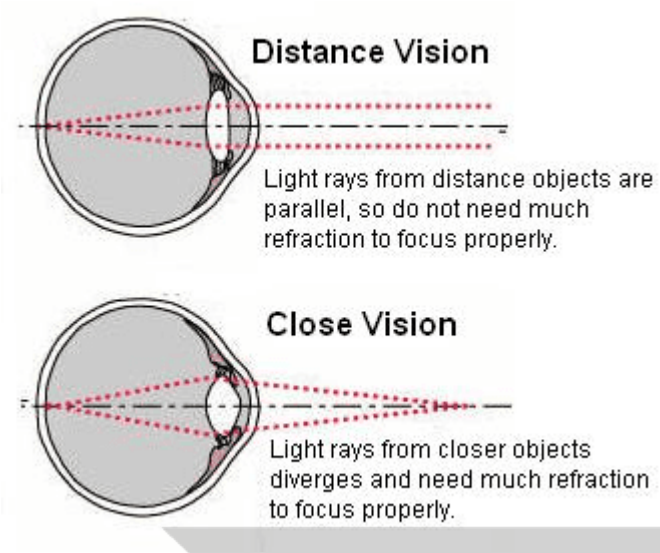



**Example Past Year Question**

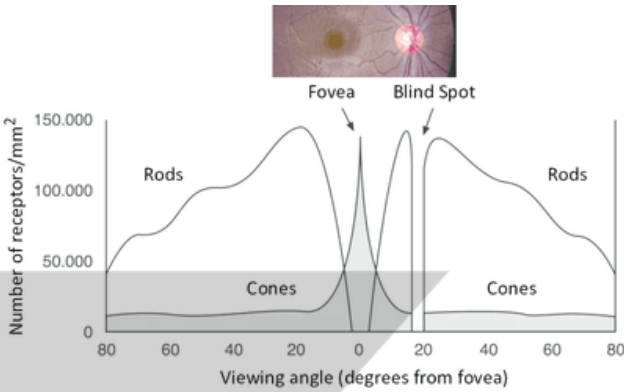
Explain why a person cannot focus on distant objects if the suspensory ligaments become permanently overstretched. (0610/42/F/M/23)

1. ciliary muscles relax
2. suspensory ligaments can no longer become tight
3. the lens is not stretched/remains wide
4. the angle of refraction remains unchanged

## 1.10. Rods and Cones

| Rods | Cones |
|---|---|
| Provide low detail, black & white images, suitable for seeing in low-intensity light (at night). | Provide detailed, coloured images; they work in high light intensity. |
| Packed most tightly around the edge of the retina, so you can see things most clearly when not looking directly at them. | Most tightly packed at the retina's centre, objects are seen most clearly when directly looking at them. |

- Fovea:
  - Part of the retina where the receptor cells are pushed most closely together
  - Where light is focused when you look straight at an object
- Distribution of Rods and Cones
  - More rod cells than cone cells
  - A number of cone cells peaked in the middle, showing the location of the fovea. No rod cells are present in the peak.
  - No rod and cone cells at the blind spot, as the optic nerve is located there. \n




## 1.11. Hormones

**Hormones:** A chemical substance produced by a gland and carried by the blood, altering the activity of one or more specific target organs.

### Endocrine Glands

- adrenal glands and adrenaline
- pancreas and insulin
- testes and testosterone
- ovaries and oestrogen

### Adrenaline

- A hormone secreted by the adrenal gland.
- It increases pulse rate, heart rate and pupil diameter.
- Increases blood glucose concentration for respiration.
- Adrenaline is secreted, for example, bungee jumping or riding a rollercoaster.

| Gland | Hormone | Function |
|---|---|---|
| Adrenal gland | Adrenaline | Prepares the body for vigorous action |
| Pancreas | Insulin | Reduces the concentration of glucose in the blood |
| Testes | Testosterone | Causes the development of male sexual characteristics |
| Ovary | Oestrogen | Causes the development of female sexual characteristics |
| Pancreas | Glucagon | Increases concentration of glucose in the blood |

## 1.12. Homeostasis

**Homeostasis:** The maintenance of a constant internal environment.

- Insulin decreases blood glucose concentration.



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- The concept of homeostatic control by negative feedback with reference to a set point

### Negative Feedback Concept

Negative Feedback: controls the production of hormones and regulates their own production

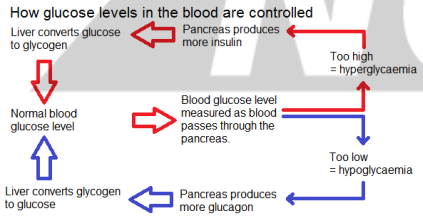
- A negative feedback control is when the change in hormone level acts as a signal to cancel out that change, so when the blood hormone level is low, hormone production is stimulated; when it is high, it is inhibited.

## 1.13. Nervous and Hormonal Control

| Comparison | Nervous system | Endocrine system |
|---|---|---|
| Speed of action | Very rapid | Can be slow |
| Nature of message | Electrical impulses travelling along nerves | Chemical messengers (hormones) travelling in the bloodstream |
| Duration of response | Usually within seconds | It may take years (puberty) |
| Area of response | Localized response (only one area usually) | Widespread response (in many organs) |
| Example of process-controlled | Reflexes such as blinking | Development of the reproductive system |

## 1.14. Homeostasis: Blood Glucose Control

- Blood glucose levels are monitored and controlled by the pancreas
- The pancreas produces and releases different hormones depending on the blood glucose level
- Insulin is released when blood glucose levels are high – the liver stores excess glucose as glycogen
- Glucagon is released when blood glucose levels are low – the liver converts stored glycogen into glucose and releases it into the blood




- When the control of blood glucose does not work, a person is said to have diabetes

### Type 1 Diabetes

Type 1 Diabetes: caused by the death of the cells that secrete insulin.

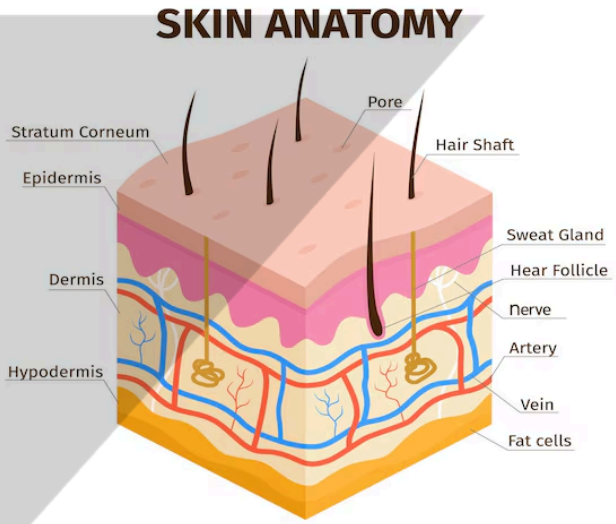
- Symptoms: hyperglycaemia (feeling unwell, dry mouth, blurred vision, and thirsty) or hypoglycaemia (tired, showing confusion and irrational behaviour)
- Treatment: eating little and often and avoiding large amounts of carbohydrates, injecting insulin to reduce blood glucose concentration

## 1.15. Homeostasis: Temperature Regulation

Control of body temperature is a homeostatic mechanism

- The human body maintains the temperature at which enzymes work best, around 37°C
- If body temperature increases over this temperature, enzymes will denature and become less effective at catalysing reactions such as respiration

### The Structure of the Skin




### Temperature Regulation by The Skin

- Regulation is controlled by the brain which contains thermoreceptors (sensitive to the temperature of the blood)
- The skin also has thermoreceptors and sends nervous impulses to the brain via sensory neurones
- The brain responds to this information by sending nerve impulses to the motor neurones, then to the effectors in the skin to maintain the temperature within a narrow range of the optimum, 37°C
- Fatty tissue under the dermis acts as a layer of insulation to prevent too much body heat from being lost through the skin



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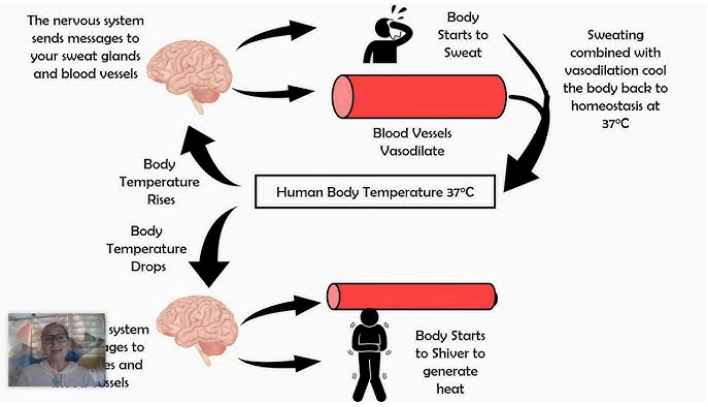


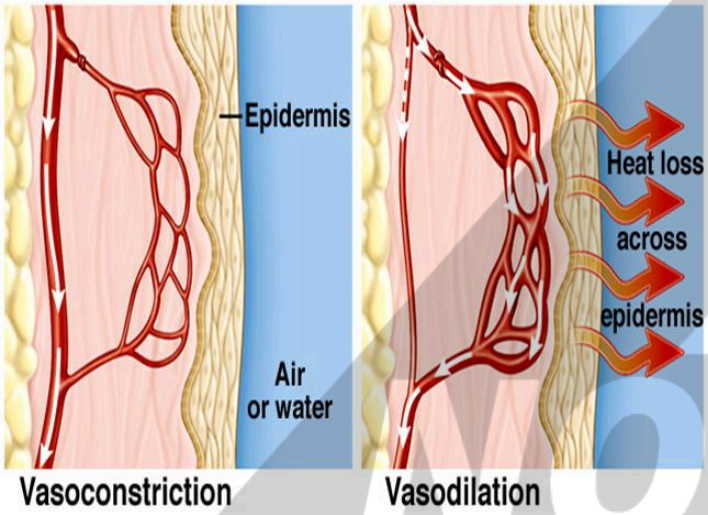

### Vasoconstriction and Vasodilation

**Vasoconstriction**: blood flow in capillaries slows down because arterioles leading to the skin capillaries get narrower when we are **cold**.

- This reduces the amount of heat lost from the blood by radiation as less blood flows through the surface of the skin

**Vasodilation**: blood flow in capillaries increases because blood vessels to the skin capillaries get wider when we are **hot**.

- This cools the body as blood (which carries heat around the body) flows at a faster rate through the skin's surface and so more heat is lost by radiation




## 1.16. Tropic Responses

- **Auxin:**
  - Plant hormones or growth substances.
  - Controls tropisms
  - It is produced by cells at the tip of the roots and shoots of plants

**Gravitropism:** a response in which a plant grows towards (positive) or away (negative) from gravity.

- **Auxins' role in gravitropism:**
  - Made in the shoot tip
  - Then, it diffuses through the plant from the shoot tip
  - Auxin is unequally distributed in response to light and gravity
  - Auxin stimulates cell elongation

**Phototropism:** a response in which a plant grows towards (positive) or away (negative) from the direction light is coming.

- **Auxins' role in phototropism:**
  - If the sun shines on the right side of a plant's shoot, auxins accumulate on the dark opposite left side.
  - Auxins accumulating makes cells on the left side grow faster than cells on the right.
  - When the left side of the shoot starts growing faster than the right side, the shoot will start to bend to the right side towards sunlight.



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